Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Concept of Operations

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1.0 Introduction

1.1 Project Description

1.1.1 Background

Purpose: The System lets an operator on the ground command, monitor, and update a controller in space that nobody can reach, and keeps that controller working between contacts.

Need: A controller that nobody can reach must stay operable, observable, and correctable for its whole life. Operation, observability, and maintenance including in-flight modification over the specified lifetime are three areas a space software specification must cover [3, Annex B, items 5.11–5.13, p. 113].

Worst-Case Failure: A failed update can leave the controller unable to run any firmware and unable to accept new firmware. Without physical access, that state is permanent.

Rationale: The loss has happened. NASA records a mission terminated after an in-flight software upload omitted a patch for a defective countdown timer; its recommendations include test-bed verification of in-flight software updates [4, Lesson 0559, p. 199].

1.1.2 Assumptions and Constraints

C-1 The ground cannot talk to the controller directly. Every command, report, and firmware image passes through the onboard computer.

C-2 All software is built to NASA's software engineering requirements. Rigor per part is set by its software class [5, Sec. 3.5.1, p. 24; App. C].

C-3 One engineer builds and checks the System. No independent team reviews the work.

C-4 The System runs on commercial development boards, not on spacecraft hardware. It can show how the System behaves, not that it would survive launch or space.

A-1 Once launched, physical access to the onboard computer or controller is impossible; all fixes and updates must arrive by radio.

A-2 The controller can lose power at any moment, even halfway through writing new firmware. The System must survive that without losing the controller [2, App. A, Table 2, p. 63].

A-3 The ground can reach the spacecraft only during line-of-sight contact windows [**TBD-4**]. Between contacts the System must work on its own.

A-4 One person operates the ground station. No second person checks a command before it is sent, so the System itself must refuse wrong commands.

A-5 Radiation in space can flip single bits in memory, so stored firmware can change on its own. The System must not run firmware that has changed [2, App. A, Table 2, p. 63].

Note. *A constraint (C) is a limit the System must work within; it is imposed, not chosen. An assumption (A) is a condition taken as true without proof; the design depends on it, and if it proves false the design may be wrong.* ***[TBD-****n****]*** *marks a value not yet known; all are listed in App. C.*

1.2 Overview of the Envisioned System

1.2.1 Overview

The System comprises two spatially separated operational segments. The **ground segment** on Earth includes the operator, the mission operations center running ground software, and the ground station antenna [6, Sec. 11.2, p. 305]. The **space segment** in orbit consists of the onboard computer and the controller, which operate without physical access.

Communication between segments is strictly established via a radio link during line-of-sight contact windows [**TBD-4**]. Outside contact windows, the space segment operates autonomously. Within the spacecraft, the onboard computer communicates with the controller over the data link during normal operation. If the controller's main firmware fails to execute, the onboard computer utilizes the reprogramming link to restore functional firmware independently of the controller's primary software. Figure 1 illustrates this segment layout.

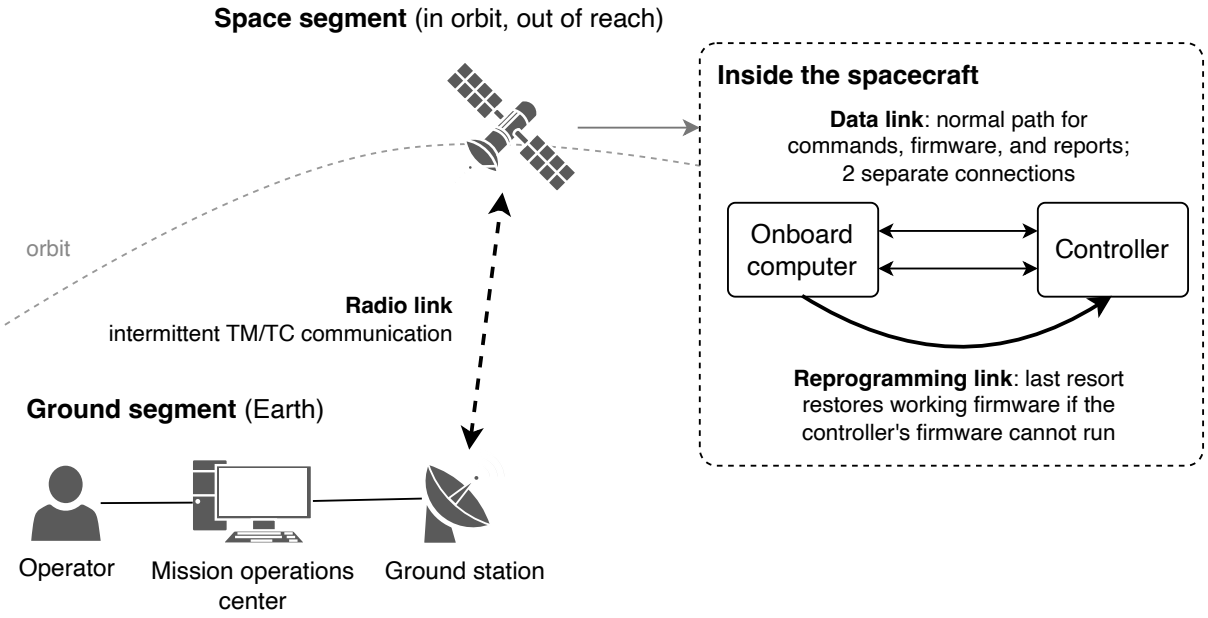


Figure 1: Ground and space segments, joined only by the radio link during contact windows. Inside the spacecraft, the data link handles normal communication, while the reprogramming link restores working firmware if the controller's firmware cannot run.

1.2.2 System Scope

The system context defines the boundary of the System: the system of interest, its users, and the external elements that interface with it [7, Sec. 8.3, p. 24].

Context Element	System Boundary
System of Interest	Onboard computer, controller, ground software in the MOC
Users	Ground operator
External Elements	Radio equipment, spacecraft power supply

- **Enabling systems:** firmware build tools and bench test setups required for development and verification, external to operational scope.
- **In scope:** commanding, telemetry, firmware updates, and autonomous fault handling between contact windows.
- **Out of scope:** onboard computer software updates, and the radio equipment, which is used as given.

2.0 Documents

2.1 Applicable Documents

[1] NASA, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, 2016. This outline follows its App. S, pp. 251–253, tailored per its Sec. 3.11.1, p. 34: App. C added for open items per its App. C, pp. 197–198.

2.2 Reference Documents

[2] NASA, *Software Assurance and Software Safety Standard*, NASA-STD-8739.8B, 2022. Off-nominal scenarios cite App. A, Table 2.
[3] ESA, *Space engineering — Software*, ECSS-E-ST-40C Rev. 1, 30 April 2025.
[4] NASA, *Fault Management Handbook*, NASA-HDBK-1002, Draft 2, 2 April 2012. A draft, not a released standard.
[5] NASA, *NASA Software Engineering Requirements*, NPR 7150.2D.
[6] NASA, *State-of-the-Art of Small Spacecraft Technology*, Chapter 11, "Ground Data Systems and Mission Operations", 2026 edition. Ground system architecture in Sec. 11.2 and Fig. 11.2, pp. 305–306.
[7] NASA, *NASA Systems Modeling Handbook for Systems Engineering*, NASA-HDBK-1009A, 12 March 2025.

3.0 Description of Envisioned System

3.1 Needs, Goals and Objectives

NEED A controller that nobody can reach must stay operable, observable and correctable for its whole life.

G-1 **Operability:** the controller keeps running and acts on the operator's commands.
- **O-1** Every valid operator command is executed. Valid means from the operator, allowed in the current state, with possible values, and not expired.
- **O-2** Firmware and commands not from the operator are never acted on.
- **O-3** With one data-link connection lost, commands, telemetry and updates continue.
- **O-4** No fault found between contacts goes unreported.

G-2 **Observability:** the ground can know what the System is doing, how healthy it is, and what happened between contacts.
- **O-5** No report is lost without the ground being told.

G-3 **Correctability:** the controller's firmware can be replaced for its whole life without losing the controller.
- **O-6** No update, even one interrupted by a single fault, leaves the controller without runnable firmware.

3.2 Overview of System and Key Elements

Operator The user. Decides and sends commands, releases new firmware, and reads the reports.

Ground software In the mission operations center. Turns the operator's decisions into commands and firmware ready to send, checks them before they leave, and keeps every report received.

Onboard computer Checks commands and firmware and passes them to the controller, stores the controller's reports between contacts, watches the controller, and restores it when its firmware cannot run.

Controller Runs its firmware, acts on commands, reports its state, and returns to working firmware on its own when new firmware fails.

The ground station's radio equipment and the spacecraft power supply are outside the System; it uses them as given.

3.3 Interfaces

External interfaces

Operator interface Human. Ground software to operator. Allows the flight operator to issue telecommands, authorize firmware deployments, and monitor system telemetry.

Radio link Radio frequency. Ground software to onboard computer via external radio equipment. Provides intermittent telemetry and telecommand (TM/TC) transfer exclusively during contact windows.

Power Electrical. Spacecraft power subsystem to onboard computer and controller. Subject to unannounced power interruptions at any time (assumption A-2).

Interfaces between elements

Data link Digital data. Onboard computer to controller. Nominal interface for commands, telemetry, and firmware updates while controller firmware is running. Uses two separate connections, so a single connection failure does not interrupt operations.

Reprogramming link Digital data. Onboard computer to controller. Recovery interface used as a last resort to restore functional firmware independently of the controller's main software.

3.4 Modes of Operations

Life-cycle modes of the System

Test Ground bench execution with simulated radio link degradation and intentional fault injection to demonstrate operational scenarios.

Operation Active mission execution using the controller modes detailed below.

Disposal Not applicable; limited to bench demonstration scope.

Controller modes during operation

OPERATE Nominal state executing confirmed firmware. New firmware images can be staged in the background during execution.

TRIAL Entered upon transitioning to updated firmware. Executes the new image for a bounded evaluation window [**TBD-5**]. If the new image is judged good, the update is committed and the controller returns to OPERATE; if not, it enters FALLBACK.

FALLBACK Entered following a failed firmware trial or boot failure. Reverts execution to the previous known-good firmware image or the last-resort firmware.

RECOVERY Entered when no operational firmware image can be executed. Controller software execution is halted while the onboard computer restores functional firmware via the reprogramming link.

3.5 Proposed Capabilities

Each capability is shown in detail by the scenarios in Section 6.0.

Capability-1 **Telemetry** (serves G-2): regular and event health reports, time-tagged, stored between contacts and sent down at the next one, with any loss reported.

Capability-2 **Command handling** (serves G-1): accept only the operator's valid telecommands, answer and log every one, and support arming and time-tagged execution.

Capability-3 **Fault handling** (serves G-1): detect faults between contacts, respond without the ground, and record them.

Capability-4 **Update** (serves G-3): receive, check, install and judge new firmware, and return to working firmware without outside help if it fails [**TBD-2**] [**TBD-3**].

Capability-5 **Test** (verification only): degrade the radio link; cut power or break a data-link connection on command.

Disposal None; bench demonstration.

4.0 Physical Environment

This section specifies the environment the System encounters across its life cycle, encompassing simulated mission operations and laboratory integration and testing. For each environmental condition, the expected system response is defined as nominal operation, degraded performance tolerance, or survivability [1, App. S, p. 252].

Phase	Environmental condition	Response	Shown on bench
Operation	Thermal range [**TBD-8**]	OPERATE	NO
Operation	Vacuum	OPERATE	NO
Operation	Radiation: single-event upset / memory bit errors	SURVIVE	SIMULATED
Operation	Electrical: power interruption at any instant	SURVIVE	YES
Launch	Mechanical: vibration and launch loads	OPERATE	NO
Integration and Test	Laboratory bench: ambient room temperature and pressure	OPERATE	YES

Legend: response

OPERATE All functions continue normally.

TOLERATE Partial functional degradation occurs, but no function is permanently lost.

SURVIVE Operational functions temporarily halt and automatically recover upon removal of the condition.

Legend: shown on bench

YES The condition itself is reproduced on the bench.

SIMULATED The condition's effect is injected on purpose; the condition itself is not reproduced.

NO Beyond what bench hardware can show (constraint C-4).

Transportation: not applicable; system transportation is outside the scope of bench demonstration.

5.0 Support Environment

Operational planning The ground operator plans contact windows in advance, defining telecommand queues, firmware uploads, and telemetry retrieval profiles, with schedule timing derived from orbital parameters [**TBD-4**].

Commanding and uploads Ground software formats and validates operator telecommands for real-time execution or time-tagged scheduling, while verified firmware

	images are released from the ground segment and uploaded across one or multiple contact windows.
Maintenance and repair	Physical maintenance is excluded, as there is no physical access in orbit (assumption A-1); anomalies are managed by autonomous onboard fault protection or remotely through operator telecommands and firmware updates.
Sparing	Physical hardware spares are not available; the onboard computer retains a known-good firmware image and the controller maintains its last-resort firmware.
Upgrades	Controller firmware updates constitute the sole operational upgrade mechanism, governed by a defined release cadence [**TBD-7**], while the onboard computer's baseline software remains fixed.
Continued support	Engineering sustainment support remains allocated for firmware build, verification, and release throughout the active mission life cycle.

6.0 Operational Scenarios

Each scenario follows one thread of events from trigger to outcome [1, App. S, pp. 252–253]. Every scenario below is demonstrated on hardware and recorded. Sources in brackets are items of NASA's list of software causes of hazards [2, App. A, Table 2].

6.1 Nominal Conditions

Scenario-100 Firmware update.

1. The operator releases new firmware; the ground software checks it and prepares the upload.
2. Over one or more contacts, the firmware goes up the radio link to the onboard computer.
3. The onboard computer checks that it is undamaged, from the operator, built for this controller, and not older than the minimum version.
4. The onboard computer delivers it to the controller over the data link; the controller keeps working meanwhile.
5. The controller switches to the new firmware and enters TRIAL [**TBD-2**].
6. The new firmware is judged good within the trial time [**TBD-5**] [**TBD-3**]; the update is committed and the controller returns to OPERATE.
7. The outcome is stored and sent down at the next contact.

Scenario-110 Telemetry between contacts.

1. The controller sends regular health reports, and an event report as soon as something important happens.
2. The onboard computer time-tags and stores every report.
3. At the next contact, all stored reports go down the radio link.

Scenario-120 Command during a contact.

1. The operator decides a command; the ground software prepares and checks it.
2. During a contact, the command goes up the radio link.
3. The onboard computer checks that it is from the operator, new, allowed in the current mode, and has possible values. A dangerous command must first be armed by a separate command; the arm expires if the command does not follow within a set time.

4. The command is executed. The answers (received, then done with the result) go down, and the command and its answers are logged.

Scenario-130 **Scheduled command.**

1. During a contact, the operator sends a command with a run time; the System answers "received".
2. At the run time, the command runs.
3. The "done" answer is stored and sent down at the next contact.

6.2 Off-Nominal Conditions

The off-nominal scenarios cover low performance, operator errors, failures, and unexpected environmental conditions [1, App. S, p. 253]. Each entry gives the trigger, then the expected outcome.

Low performance of the radio link

Scenario-200 The radio link is lost during a firmware upload. The upload resumes at the next contact from where it stopped.

Scenario-210 A long gap between contacts fills the report storage on the onboard computer. From then on, the System drops the oldest regular health reports to make room and keeps every event report. At the next contact, it sends what it stored and tells the ground how many health reports were dropped and the time span they covered.

Operator errors and unauthorized commands

Scenario-300 A command not from the operator, or an old command sent again, arrives. It is refused and logged [2, p. 62].

Scenario-310 A command not allowed in the current mode arrives, for example "switch" when no new firmware has arrived. It is refused with the reason [2, p. 63].

Scenario-320 A command with impossible values arrives. It is refused with the reason [2, p. 62].

Scenario-330 A dangerous command arrives without being armed. It is refused with the reason.

Scenario-335 A dangerous command arrives after its arm has expired. It is refused with the reason; the operator must arm again.

Scenario-340 A command arrives after its expiry. It is refused with the reason [2, p. 63].

Scenario-350 An answer is lost and the operator sends the command again. It is not run a second time; the answer is "already done".

Scenario-360 Firmware built for a different controller is uploaded. It is refused before it is installed [2, p. 63].

Firmware update failures

Scenario-400 Firmware is damaged in transfer. The onboard computer refuses it before it reaches the controller [2, p. 63].

Scenario-410 Firmware not released by the operator arrives. It is refused.

Scenario-420 Firmware older than the minimum version arrives. It is refused.

Scenario-430 The new firmware hangs during TRIAL. The controller returns to its previous firmware without outside help.

Scenario-440 The new firmware restarts over and over. After **[TBD-6]** failed starts, the controller returns to its previous firmware [2, p. 64].

Scenario-450 The new firmware runs but is never judged good. After the trial time, the controller returns to its previous firmware.

Scenario-460 The previous firmware is also unusable. The controller runs its last-resort firmware.

Scenario-470 The operator cancels an update in progress. The update stops; working firmware is left running.

Scenario-480 No firmware can run. The onboard computer restores the controller through the reprogramming link [2, p. 63].

Equipment failures

Scenario-500 One data-link connection fails. Everything continues on the other, and the failure is reported.

Scenario-510 Both data-link connections fail. The controller keeps working on its current firmware, the loss is reported, and the reprogramming link is still usable.

Scenario-520 The controller goes silent. The onboard computer notices, tries to bring it back, and records what it did.

Scenario-530 A program on the onboard computer crashes. It restarts and resumes any update in progress; the controller is not affected.

Unexpected environmental conditions (from Section 4.0)

Scenario-600 Power is lost while the controller's firmware is being written. When power returns, the controller runs its previous firmware, and the update can be repeated [2, p. 63].

Scenario-610 A radiation bit error changes stored firmware. The controller detects it before starting that firmware and does not run it [2, p. 63].

Scenario-620 A radiation bit error hits the memory the controller is running from. The controller may crash; it restarts on its own and resumes operation [2, p. 63].

7.0 Impact Considerations

7.1 Environmental Impacts

None. A bench demonstration creates no orbital debris, contamination or hazardous waste.

7.2 Organizational Impacts

One engineer builds the System and supports it for its whole life. The operator must know the operating procedures: arming dangerous commands, scheduling commands, and reading the reports.

7.3 Technical Impacts

The System shows, with a recorded result for every scenario, that a controller nobody can reach can be updated, commanded and monitored without being lost. The service it provides is fail-safe in-orbit reprogramming of spacecraft controllers.

8.0 Risks and Potential Issues

R-1 Operations: both data-link connections may share software at each end. One defect could cut both at once, leaving only the reprogramming link.

R-2 Implementation approach: code drafted with an AI assistant may contain plausible but wrong behaviour. Each subsystem is reviewed and tested before it is accepted.

R-3 Staffing: one engineer builds and supports the System. Losing that person stops new firmware for the rest of the mission.

Closeout: at the end of the project, the documents and the recorded result of every scenario stay in the repository.

A Acronyms

ConOps Concept of Operations
MOC Mission operations center
TBD To Be Determined
TC Telecommand
TM Telemetry

B Glossary

Confirmed Firmware judged good after its trial; only confirmed firmware is known-good.
Last-resort Known-good firmware that updates never replace (golden image).
Switch The controller stops running old firmware and starts the new.
Arm First step of a dangerous command; changes nothing alone and expires after a set time.
Expiry Time after which a waiting command is refused.

C Open Items

All statements in this document were stated or accepted by the author on 9–10 October 2026. The values below are not yet known.

TBD-1 Project name and document reference.
TBD-2 Switch automatically on arrival, or only on operator command.
TBD-3 Who judges new firmware good: controller or operator.
TBD-4 Contact window length and gap (sourced, or a test parameter).
TBD-5 Trial time limit (requirements).
TBD-6 Failed starts before going back (requirements).
TBD-7 Update rate.
TBD-8 Temperature range.